

How Far Off the Recorded Path Can a Gaussian-Splat Driving Simulator Be Trusted?

CS 566 Project Proposal (one document per team) · Kyle Sha (ID 9085525476) · Allen Xia (ID 9085517234) · 29 Sep 2026

Main papers: [1] **Difix3D+** (CVPR 2025 Oral), *re-implemented* · [2] **FaithFusion** (CVPR 2026), *compared* · [3] **HUGSIM** (TPAMI 2026), *simulator for evaluation*

1. Problem Statement

Driving software is increasingly evaluated in simulators reconstructed from real driving logs. The recorded video is converted into a **3D Gaussian splat** [13], a scene representation composed of millions of small anisotropic Gaussians that can be rendered from arbitrary camera poses, so that the simulated vehicle may follow a trajectory different from the recorded one. However, **a reconstruction is reliable only near the recorded trajectory**. When the simulated vehicle changes lanes or the camera is mounted higher (e.g., on a truck), rendered images degrade, and a driving system evaluated on them may behave differently than it would in reality. This project addresses two questions: **how far from the recorded trajectory a splat-based simulator remains trustworthy, and whether current generative repair methods ("fixers") restore trustworthiness or merely improve visual realism**. We define **trust** as the agreement between the outputs of a lane detector and a vehicle detector on a rendered image and on the **real** image captured at the same pose.

2. Significance and Motivation

Simulation is a principal means of validating driving software at scale. If renders away from the recorded trajectory are unreliable, a simulator may accept unsafe behaviour or reject safe behaviour, which undermines its value for validation. Establishing the range over which such simulators can be trusted is therefore of direct practical importance. Both authors are members of Wisconsin Autonomous, a student autonomous-vehicle team, where we develop and test driving software in conventional simulators (MATLAB/Simulink, RoadRunner, CARLA). These rely on hand-built 3D assets and differ visibly from real camera imagery, which limits how far their results transfer to the vehicle. Reconstruction-based simulators promise to close this gap, but only if their renders remain faithful away from the recorded trajectory; a generative fixer can make an image more convincing while making it less accurate. Characterizing this trade-off is the central motivation of the project, which draws on camera geometry, 3D reconstruction, image restoration, and CNN-based perception.

3. State of the Art

- **Splat-based simulators:** HUGSIM [3] executes driving policies in reconstructed scenes; the EUVS benchmark [7] shows that current methods degrade under large viewpoint changes.
- **Diffusion-based fixers:** Difix3D+ [1] repairs off-trajectory renders with a single-step diffusion model and distills the repaired views back into the 3D scene, reporting an approximately $2\times$ FID improvement over prior methods. NVIDIA Fixer [4] applies the same approach to driving data, and FaithFusion [2] restricts how much a fixer may overwrite the reconstruction using a per-pixel confidence map.
- **Evaluation practice:** these methods are evaluated primarily with image-realism metrics (FID, LPIPS) or with shifted cameras for which no real image exists (PRG-Fusion [6]). Studies with offset analysis or real off-trajectory images [9, 10, 11] neither compare fixers nor quantify hallucinated content. **To our knowledge, no prior work compares fixers against real off-trajectory images in terms of the content they invent.**

4. Proposed Approach

- **Re-implementation:** we re-implement Difix3D+ [1], comprising the fixer's training procedure and its progressive update that distills repaired views into the 3D scene. The implementation is written from the paper on the same pretrained diffusion model used by the authors, validated against NVIDIA's official code and weights, and then trained on driving data.
- **New contributions:** (a) an **evaluation against real ground truth**, using Para-Lane [8], which scans the same roads in three parallel lanes (approximately 3 m and 6–7.7 m apart), and MV2 [9], in which a car and a scooter record the same road at different camera heights; training pairs are likewise constructed from neighbouring real lanes (cf. ReCamDriving [12]); (b) **hallucination accounting**, measuring invented or erased lane markings and spurious vehicles in addition to image similarity; (c) a **lightweight fixer** trained from scratch (15M parameters, about $60\times$ smaller than the Difix backbone); (d) a **geometry-only road-surface prior**, adapted from ViSE [15], which regularizes the road using LiDAR and cannot introduce new content; (e) a **cross-dataset transfer study** (EUVS/MARS [7], MV2 [9]); and (f) **coverage-gated faithful fixing**, a new method that restricts a fixer's edits to regions the reconstruction cannot support: the output blends the render and the fixer's prediction with a per-pixel gate derived from splat coverage ($1 - \alpha$), the same gate weights the progressive-distillation loss, and an object-consistency penalty (a frozen detector flagging vehicles absent from the source view) discourages spurious vehicles. Unlike FaithFusion's information-gain map, the gate requires no additional rendering pass; it is motivated by our

preliminary finding that coverage modestly predicts incorrect fixer edits, better than information gain in all eight method–offset pairs (AUROC 0.61–0.77 versus 0.53–0.67), though not invented lane markings, which arise in well-covered regions.

5. Limitations of Existing Approaches and Expected Advantages

Existing fixers are diffusion models trained to produce realistic images; they are therefore rewarded for plausible detail even where that detail is incorrect. A preliminary study (September 27–28) is consistent with this: at approximately 3 m lateral offset, lane-detector agreement with the real image falls from 0.98 to 0.07, and NVIDIA’s released Difix improves image similarity (LPIPS 0.52 $\rightarrow$ 0.44) but not lane-marking precision (0.52 $\rightarrow$ 0.48). We test four hypotheses. **(H1)** Training a fixer on real adjacent-lane pairs makes it more faithful, not only more realistic; in the preliminary study, our re-implemented Difix raised lane agreement from 0.07 to 0.71 on two held-out roads, at the cost of more spurious vehicles (0.18 versus 0.04 per image). **(H2)** A geometry-only prior improves lane fidelity without introducing hallucinated content. **(H3)** Fixers trained on one city do not transfer to others, so trustworthiness must be established per domain. **(H4)** Coverage gating with the object-consistency penalty (f) preserves the lane-fidelity gain of H1 while reducing spurious vehicles toward the rate of the raw render.

6. Evaluation Plan and Expected Results

- **Deviation axes:** lateral offset (real images at $\approx$ 3 m and 6–7.7 m), camera height (MV2 car–scooter pairs; road-plane-warped reference as fallback), and heading (secondary; exact warp, observed regions only).
- **Metrics:** image similarity to the real image (PSNR, LPIPS); lane-marking precision and recall (invented versus erased markings); lane-detector (CLRNet [14]) and vehicle-detector (YOLO11) agreement, following the metrics of ReconDreamer [5]; spurious-vehicle count; and, as an extension, a driving policy’s score within HUGSIM [3].
- **Expected results** (all methods of §4 and the raw render): *Figure 1*, trust versus lateral offset for each method; *Figure 2*, trust versus camera height; *Table 1*, hallucination statistics per method; *Table 2*, cross-dataset transfer; *Table 3*, our Difix re-implementation versus the official code; and side-by-side videos with hallucinated content annotated.

Component	In the paper	Our version
Difix fixer [1]	Diffusion fixer; official code and weights released	Re-implemented and retrained by us on driving pairs; checked against official
Progressive update [1]	Fixed images fed back into the 3D scene	Re-implemented
FaithFusion [2], NVIDIA Fixer [4]	Released code / weights	Run as baselines
Lightweight fixer	—	New, trained from scratch
Coverage-gated fixer (f)	—	New method (gating + object-consistency fine-tuning)
Road prior [15]	Road-surface model (ViSE)	Adapted and re-implemented

7. Timeline

Dates	Kyle Sha	Allen Xia
9/29–10/12	MV2 reconstruction and camera-height evaluation; additional Para-Lane test scenes	Evaluation framework for MV2 and EUVS; per-scene variance analysis
10/13–10/29	Validation of the Difix re-implementation against the official code	Road-surface prior on all scenes; hallucination analysis; mid-term report 10/29
10/30–11/20	Cross-dataset transfer study; progressive update on all scenes	Coverage-gated fixer (f); closed-loop evaluation in HUGSIM (extension)
11/21–12/10	Final figures and videos	Presentation (12/1–12/10) and project webpage (12/10)

Risks: preliminary results rest on few test roads (more scenes and per-scene variance will be reported); closed-loop evaluation uses the LTF policy, as others require a legacy software stack. Webpage: ksha23.github.io/cs566-drive-sim.

References

1. Difix3D+, CVPR 2025 · arxiv.org/abs/2503.01774 · github.com/nv-tlabs/Difix3D
2. FaithFusion, CVPR 2026 · arxiv.org/abs/2511.21113 · github.com/wangyuanbiubiubi/FaithFusion
3. HUGSIM, TPAMI 2026 · arxiv.org/abs/2412.01718 · github.com/hy Zhou404/HUGSIM
4. NVIDIA Fixer, 2025 · github.com/nv-tlabs/Fixer
5. ReconDreamer (detector-agreement metrics), CVPR 2025 · arxiv.org/abs/2411.19548
6. PRG-Fusion, 2026 · arxiv.org/abs/2609.06948
7. EUVS benchmark, ICCV 2025 · arxiv.org/abs/2412.05256
8. Para-Lane dataset, 3DV 2025 · arxiv.org/abs/2502.15635
9. MV2 dataset, ECCV 2026 · arxiv.org/abs/2608.12442
10. From Concept to Capability (3DGS for AD), SafeComp 2026 · arxiv.org/abs/2605.01995
11. SetDiff, 2026 · arxiv.org/abs/2601.07540
12. ReCamDriving, ECCV 2026 · arxiv.org/abs/2512.03621
13. 3D Gaussian Splatting, SIGGRAPH 2023 · arxiv.org/abs/2308.04079
14. CLRNet lane detector, CVPR 2022 · arxiv.org/abs/2203.10350
15. ViSE (road-surface prior), ICCV 2025 · arxiv.org/abs/2510.18341